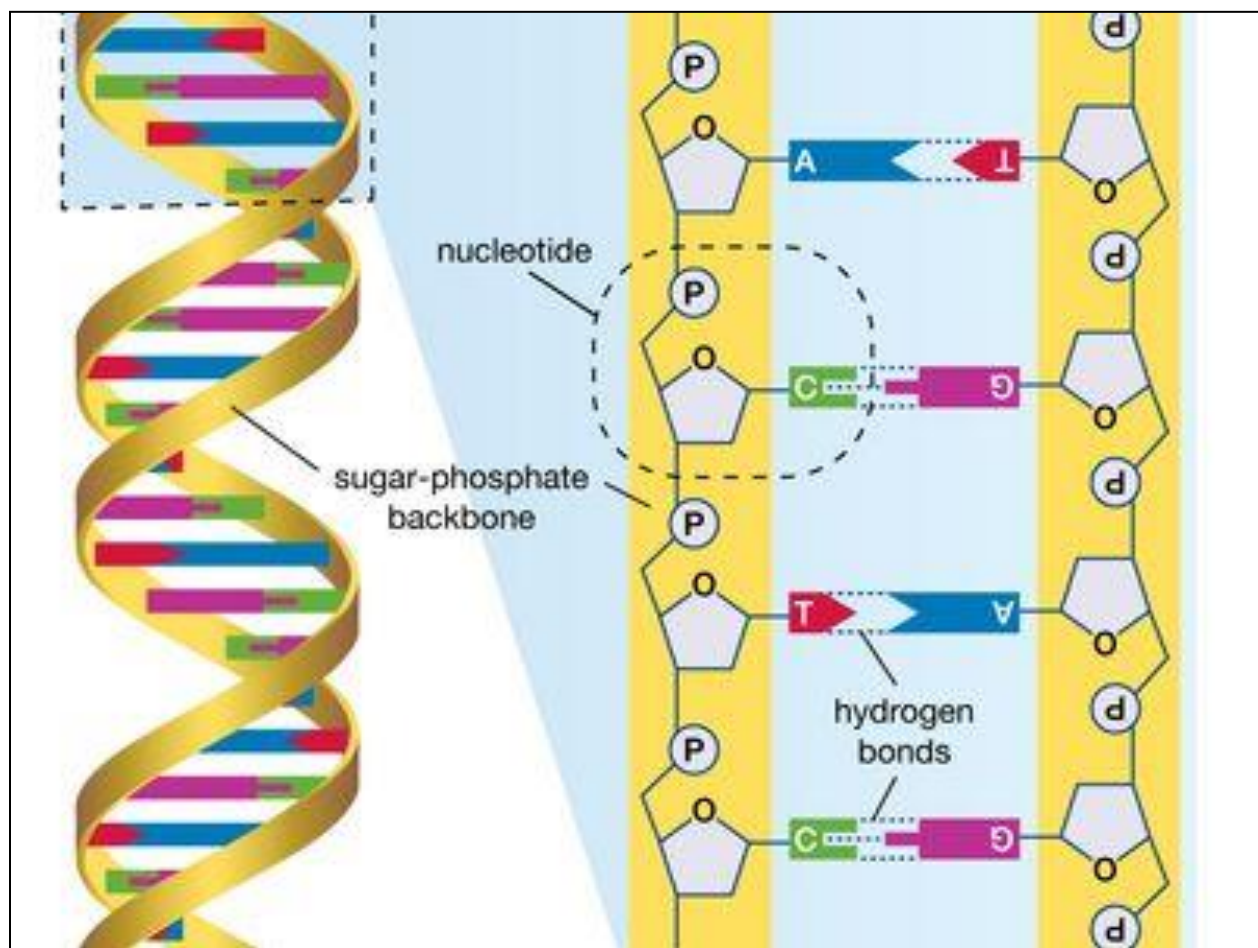
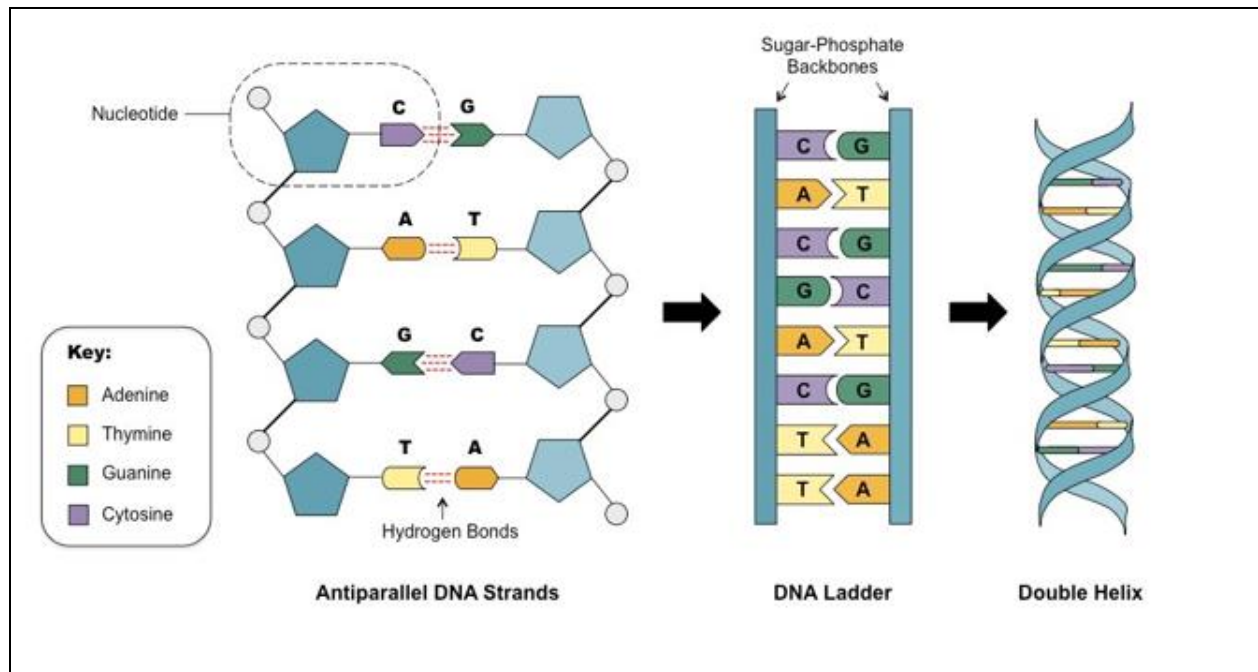


What is DNA?

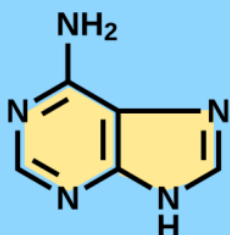
“DNA is a group of molecules that is responsible for carrying and transmitting the hereditary materials or the genetic instructions from parents to offspring.”

- ❖ DNA, or deoxyribonucleic acid, is the hereditary material in humans and almost all other organisms including plants, mammals, insects, microbes etc.
- ❖ Nearly every cell in a person's body has the same DNA.
- ❖ Most DNA is located in the cell nucleus (where it is called nuclear DNA), but a small amount of DNA can also be found in the mitochondria (where it is called mitochondrial DNA or mtDNA) [Mitochondria are structures within cells that convert the energy from food into a form that cells can use].
- ❖ The information in DNA is stored as a code made up of four chemical bases: adenine (A), guanine (G), cytosine (C), and thymine (T).
- ❖ Human DNA consists of about 3 billion bases, and more than 99 percent of those bases are the same in all people.
- ❖ The order, or sequence, of these bases determines the information available for building and maintaining an organism, similar to the way in which letters of the alphabet appear in a certain order to form words and sentences.
- ❖ DNA bases pair up with each other, A with T and C with G, to form units called base pairs.
- ❖ Each base is also attached to a sugar molecule and a phosphate molecule.
- ❖ Together, a base, sugar, and phosphate are called a nucleotide.
- ❖ Nucleotides are arranged in two long strands that form a spiral called a double helix.
- ❖ The structure of the double helix is somewhat like a ladder, with the base pairs forming the ladder's rungs and the sugar and phosphate molecules forming the vertical sidepieces of the ladder.
- ❖ An important property of DNA is that it can replicate, or make copies of itself.
- ❖ Each strand of DNA in the double helix can serve as a pattern for duplicating the sequence of bases. This is critical when cells divide because each new cell needs to have an exact copy of the DNA present in the old cell.

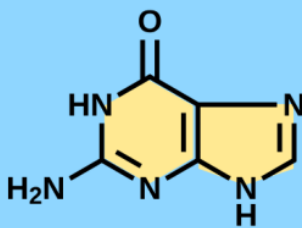


Purines and Pyrimidines

PURINES



adenine

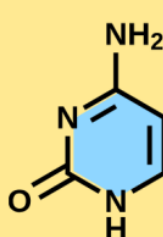


guanine

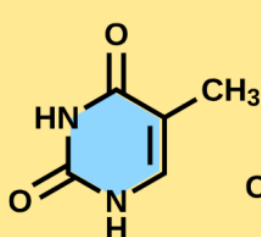
6 + 5 membered rings

Heterocyclic aromatic compound with a pyrimidine ring fused with an imidazole ring

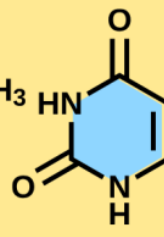
PYRIMIDINES



cytosine



thymine



uracil

6 membered ring

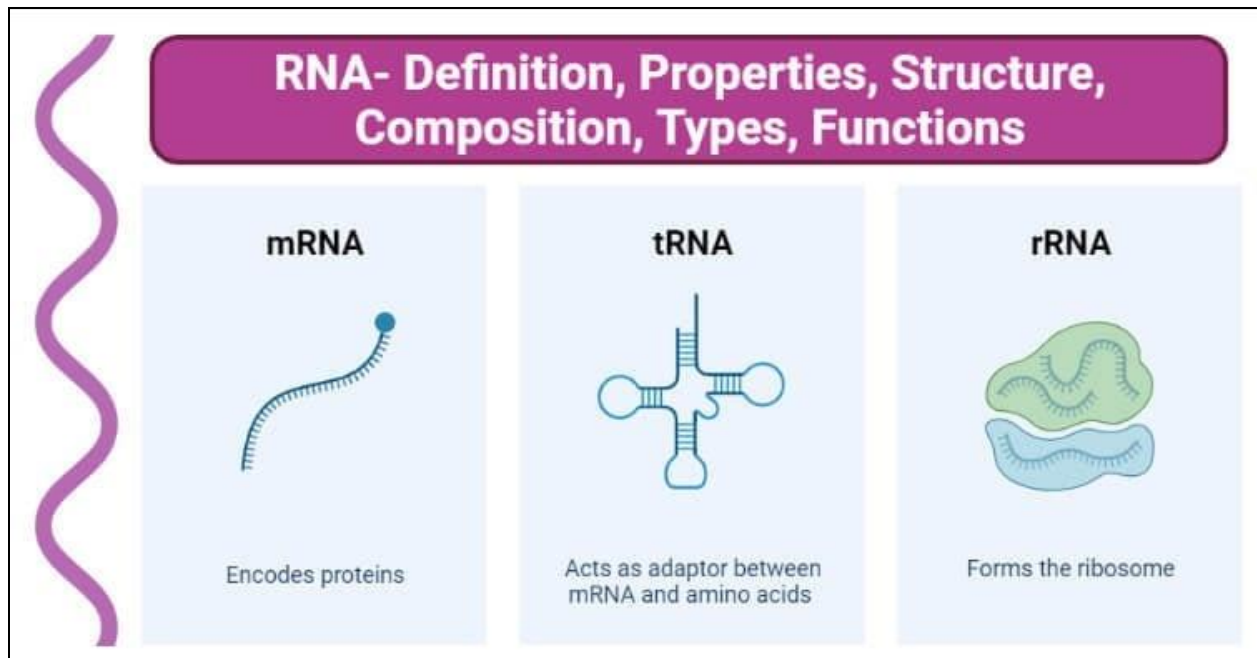
Heterocyclic aromatic compound with a pyrimidine ring, which has nitrogen atoms at the 1 and 3 positions

sciencenotes.org

What is RNA?

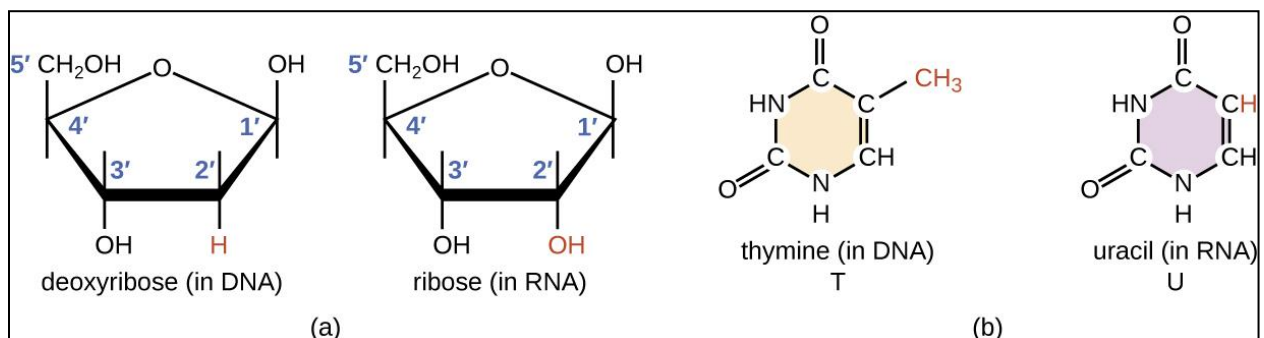
RNA (Ribonucleic acid) is a single-stranded nucleic acid molecule and made up of ribonucleotides.

- ❖ A ribose nucleotide in the chain of RNA consists of a ribose sugar, phosphate group, and a base.
- ❖ In each ribose sugar, one of the four bases is added: Adenine (A), Guanine (G), Cytosine (C), and Uracil (U).
- ❖ The base is attached to a ribose sugar with the help of a phosphodiester bond. As RNA comprises many ribose nucleotides, the length of the chains of nucleotides can vary according to their types or their functions.
- ❖ RNA thus differs from [DNA](#), on the type of sugar used to make the molecule and replacement of base Thymine in DNA with Uracil in RNA. Additionally, DNA is a double-stranded molecule whereas RNA is a single-stranded molecule.



Properties of RNA

- ❖ RNA is a single-stranded molecule and not a double helix, one of the consequences of this, is that RNA can form a variety of three-dimensional molecular complexes than DNA.
- ❖ RNA has ribose sugar in its nucleotides, rather than deoxyribose. These two sugars differ from each other in the presence or absence of an Oxygen atom.



- ❖ Ribose sugar is also a cyclical structure consisting of 5 Carbon and one Oxygen just like DNA. But the major difference is the presence of extra OH group in 2' Carbon of ribose which is absent in deoxyribose sugar.
- ❖ The OH group in 2' Carbon makes the RNA molecule prone to hydrolysis.

- ❖ Some studies have also concluded that this chemical liability of RNA due to extra OH- the group has led to DNA being the genetic storehouse as it lacks OH group in 2' Carbon making it more stable to hold information.
- ❖ RNA nucleotides carry the nitrogenous bases, Purines, and Pyrimidines. But in RNA in place of Pyrimidine Thymine, Uracil is present which too forms hydrogen bonding with Adenine just as Thymine does in DNA molecule.

Structure of RNA



Fig: Structure of RNA. Created with BioRender.com

- ❖ RNA is a typical single-stranded biopolymer of ribonucleotides bonded with each other via a phosphodiester bond.
- ❖ An RNA strand is synthesized in the 5' to 3' direction from a locally single-stranded region of DNA.
- ❖ It has ribose sugars that are attached to four bases: Adenine, Guanine, Uracil, and Cytosine. Ribose sugar has an extra OH- group in 2' Carbon as compared to deoxyribose sugar in DNA.
- ❖ This extra OH- group in RNA, has led them to be synthesized for short-term functions.
- ❖ The three-dimensional structure of RNA is critical to its stability and function.
- ❖ RNA being a single-stranded molecule can form a complex structure by allowing its ribose sugars and bases to be modified on the action of cellular enzymes (that attach chemical groups), to perform different functions.

- ❖ They are even capable of folding on themselves and showing intramolecular hydrogen bonding between complementary strands, making it a double-stranded molecule to exhibit specific function.

Composition of RNA

- ❖ RNA is a biopolymer of nucleotides bonded with each other via a phosphodiester bond.
- ❖ The nucleotide that makes up the RNA are also referred to as Ribose nucleotide due to the presence of ribose sugar in their structure. Overall, RNA is composed of a ribose sugar, phosphate, and nitrogenous base.
- ❖ Ribose sugar is a cyclical structure made up of five carbons and one oxygen atom. This sugar contains two OH-groups at 2' Carbon and 3' Carbon.
- ❖ This ribose sugar is attached to a nitrogenous base via hydrogen bonding.
- ❖ There are four nitrogenous bases namely: Adenine (A), Guanine (G), Uracil (U), and Cytosine (C).
- ❖ These nitrogenous bases pair complementarily with each other: G with C and A with U.

Types of RNA

Of many types of RNA, the three well known and most commonly discussed and found in almost all organisms. These three types of RNA are:

1. **mRNA (messenger RNA)**
2. **rRNA (ribosomal RNA)**
3. **tRNA (transfer RNA)**

1. mRNA (messenger RNA)

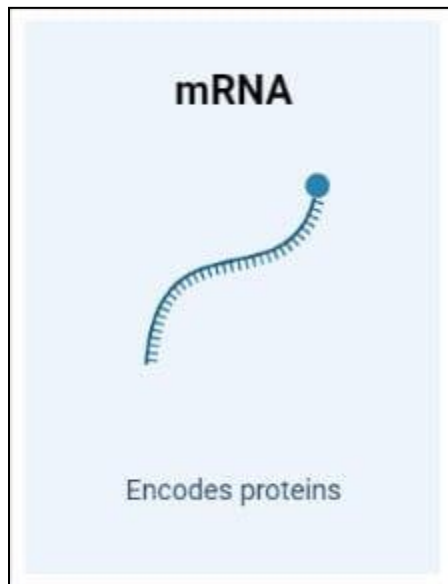


Fig. mRNA (messenger RNA). Created with BioRender.com

- ❖ It is a single-stranded RNA molecule that is complementary to one of the strands of DNA.
- ❖ mRNA is the version of the genetic materials that leave the nucleus and move to the cytoplasm where responsible proteins are synthesized.
- ❖ This RNA has utmost importance during protein synthesis, when the ribosome moves along this mRNA, it reads the base sequences and uses the genetic code to translate them into specific proteins.
- ❖ These codes are in the form of triplet sequences of nitrogenous bases and are often referred to as codons.
- ❖ As we now know that mRNA is responsible for transferring the genetic information into ribosomes where by reading the base sequences on mRNA, the translation of proteins is made possible, thus the name resembles its functions i.e., messenger RNA.
- ❖ We can even say mRNA is the molecule that uses genetic code present on a portion of DNA and make proteins. If mRNA wouldn't have existed then the information on DNA could never be used by our body.

2. rRNA (ribosomal RNA)

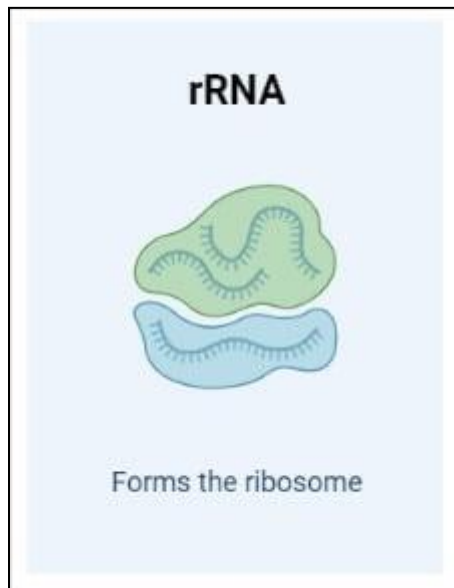


Fig: rRNA (ribosomal RNA). Created with BioRender.com

- ❖ It is a single-stranded RNA molecule found in cells that forms the part of the protein-synthesizing organelle, [Ribosome](#).
- ❖ It is synthesized inside the nucleus particularly in the nucleolus where rRNA coding genes are present. The synthesized rRNA can be of varying sizes, commonly distinguished as small and large.
- ❖ These newly synthesized rRNAs combine with ribosomal proteins and form smaller subunits and larger subunits of ribosomes respectively.
- ❖ These rRNAs are vital in recognizing conserved regions of incoming mRNAs and tRNA thus facilitating their binding and carrying out protein synthesis.
- ❖ Additionally, rRNA also has enzymatic activity (peptidyl transferase) and catalyzes the formation of the peptide bond in between two aligned proteins/amino acids during protein synthesis.

3. tRNA (transfer RNA)

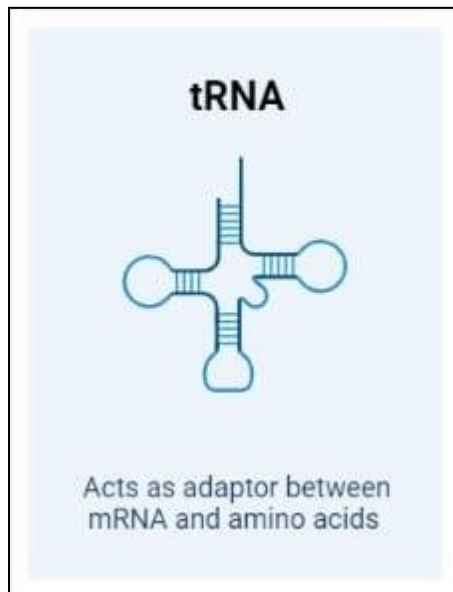
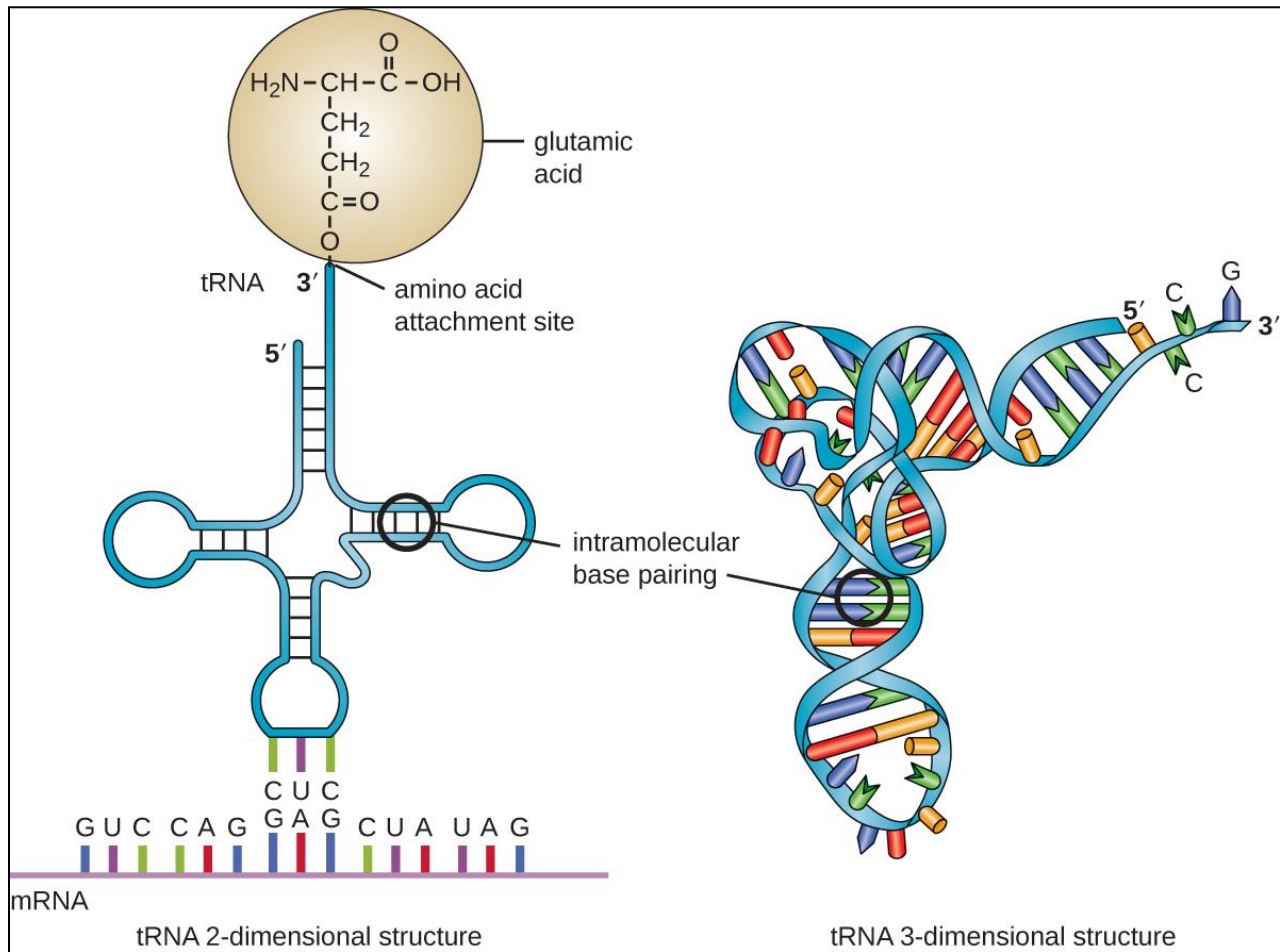


Fig: tRNA (transfer RNA). Created with BioRender.com

- ❖ It is a type of RNA molecule that helps to decode information present in mRNA sequences into specific proteins.
- ❖ It is encoded by DNA in the cell nucleus and transcribed with the help of [RNA polymerase III](#).
- ❖ The structure of tRNA folds upon itself and creates an intra complementary base pairing which gives rise to hydrogen-bonded stems and associated loops that contains nucleotides with modified bases.
- ❖ The structure in two-dimensional resembles a cloverleaf having three loops and an open end. are usually 75-90 ribonucleotides in length.
- ❖ Each of these loops consisting of arms has a distinct name and function. The three-loop consisting arms are namely: DHU or D arm, which has recognition site for specific enzyme amino-acyl tRNA synthetase; T arm that consists of ribosome recognition site and Anticodon arm that recognizes and bind to mRNA present in the ribosome.
- ❖ The open end with no loop is the site for attachment of amino acid, via 3' OH bonding with COOH- group of the [amino acid](#).
- ❖ In general, tRNA reads the code on the mRNA sequence in Ribosome and translates specific amino acid, it does so along the length of the mRNA and gives

out a polypeptide chain of amino acids (proteins) in association with other important enzymes like aminoacyl tRNA synthetase and peptidyl transferase.



Some other types of RNA

1. Ribozymes

- ❖ These types of RNAs refer to those RNAs that are capable of showing enzymatic activities.
- ❖ They were first discovered in introns of precursor ribosomal RNA of *Tetrahymena thermophilus*, where it was found that these noncoding sequences were capable of excising themselves without any protein or external source.
- ❖ These RNAs play vital roles in major reactions like RNA splicing, viral replication, and tRNA biosynthesis.

- ❖ rRNAs also show enzymatic activities and thus can be termed as Ribozymes.

2. Antisense RNAs

- ❖ Antisense RNAs are those RNAs that contain sequences that are complementary to protein-coding sequences of mRNA.
- ❖ These are single-stranded like mRNA but cannot code for proteins.
- ❖ However, they can interfere and inactivate their complementary mRNA sequences thus inhibiting protein synthesis.
- ❖ This ability of antisense RNAs has led researchers to create artificial antisense RNAs that can inhibit the protein synthesis of potential disease-causing organisms or of infected cells which then end up killing unwanted cells.

Functions of RNA

- ❖ The prime function of RNA is in protein synthesis.
- ❖ Without RNA, the information encoded in DNA could have never been transcribed to make essential proteins that a cell needs to maintain its integrity.
- ❖ mRNAs have now been widely used in pharmaceutical industries to synthesize potential vaccines.
- ❖ Moreover, mRNAs are now used to develop new categories of medicines
- ❖ mRNAs have made the formation of the cDNA library possible.
- ❖ rRNAs are structural units of Ribosomes, which are essential organelles during protein synthesis.
- ❖ Ribozymes can help suppress the expression of specific mRNA by cleaving them out without relying on the host's machinery.
- ❖ Artificial antisense RNAs are capable of arresting protein synthesis by binding with the mRNAs, which have contributed to the human's ability to combat diseases and mutations.

Functions of RNA in Protein Synthesis

Cells access the information stored in DNA by creating RNA to direct the synthesis of proteins through the process of translation. Proteins within a cell have many functions, including building cellular structures and serving as enzyme catalysts for cellular chemical reactions that give cells their specific characteristics. The three main types of RNA directly involved in protein synthesis are messenger RNA (mRNA), ribosomal RNA (rRNA), and transfer RNA (tRNA).

In 1961, French scientists François Jacob and Jacques Monod hypothesized the existence of an intermediary between DNA and its protein products, which they called messenger RNA. Evidence supporting their hypothesis was gathered soon afterwards showing that information from DNA is transmitted to the ribosome for protein synthesis using mRNA. If DNA serves as the complete library of cellular information, mRNA serves as a photocopy of specific information needed at a particular point in time that serves as the instructions to make a protein.

The mRNA carries the message from the DNA, which controls all of the cellular activities in a cell. If a cell requires a certain protein to be synthesized, the gene for this product is “turned on” and the mRNA is synthesized through the process of transcription. The mRNA then interacts with ribosomes and other cellular machinery to direct the synthesis of the protein it encodes during the process of translation. mRNA is relatively unstable and short-lived in the cell, especially in prokaryotic cells, ensuring that proteins are only made when needed.

[A diagram showing mRNA as a long strand with sets of 3 letters grouped; the left of the mRNA is labeled 3-prime, the right is labeled 5-prime. An oval labeled ribosome small subunit sits under the mRNA and spans 3 of the 3-letter groups. A larger dome (labeled ribosome large subunit) sits on top of the mRNA at this same region. The large subunit has 3 gaps where rectangles labeled tRNA sit. These rectangles each sit on a group of 3-letters on the mRNA at one end and contain an amino acid on the other end. The tRNA on the left has a single amino acid. The tRNA in the middle has a growing peptide chain

of many amino acids. The tRNA on the right as no amino acids and is leaving the ribosome.]

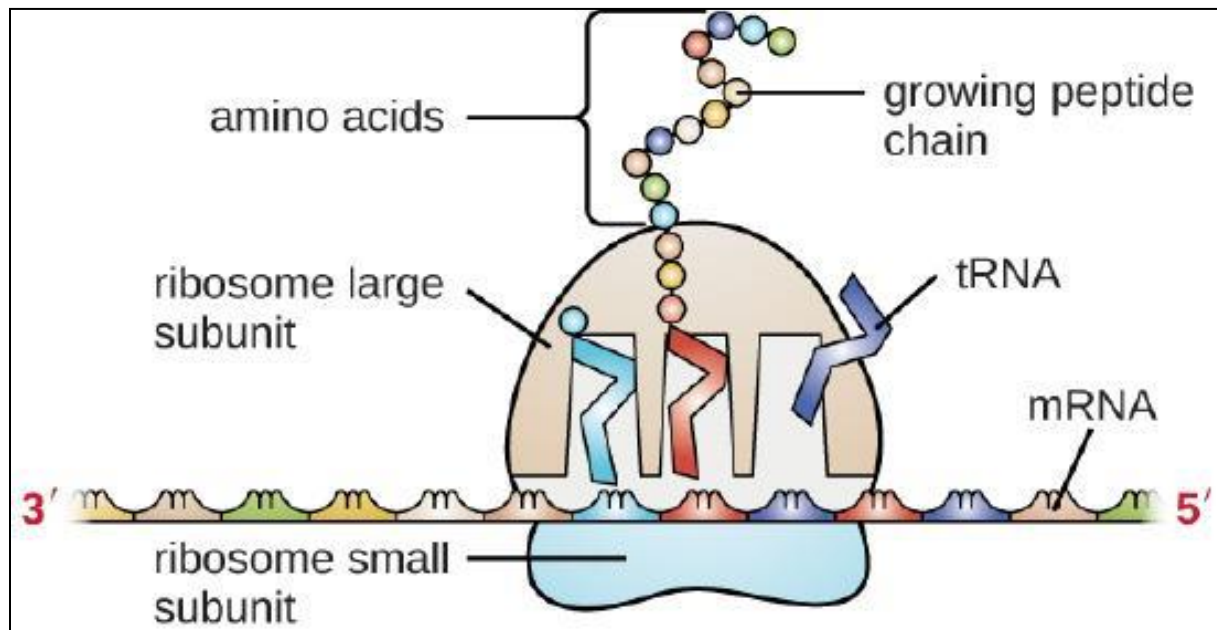


Fig: A generalized illustration of how mRNA and tRNA are used in protein synthesis within a cell.

rRNA and tRNA are stable types of RNA. In prokaryotes and eukaryotes, tRNA and rRNA are encoded in the DNA, then copied into long RNA molecules that are cut to release smaller fragments containing the individual mature RNA species. In eukaryotes, synthesis, cutting, and assembly of rRNA into ribosomes takes place in the nucleolus region of the nucleus, but these activities occur in the cytoplasm of prokaryotes. Neither of these types of RNA carries instructions to direct the synthesis of a polypeptide, but they play other important roles in protein synthesis.

Ribosomes are composed of rRNA and protein. As its name suggests, rRNA is a major constituent of ribosomes, composing up to about 60% of the ribosome by mass and providing the location where the mRNA binds. The rRNA ensures the proper alignment of the mRNA, tRNA, and the ribosomes; the rRNA of the ribosome also has an enzymatic activity (peptidyl transferase) and catalyzes the formation of the peptide bonds between

two aligned amino acids during protein synthesis. Although rRNA had long been thought to serve primarily a structural role, its catalytic role within the ribosome was proven in 2000.² Scientists in the laboratories of Thomas Steitz (1940–) and Peter Moore (1939–) at Yale University were able to crystallize the ribosome structure from *Haloarcula marismortui*, a halophilic archaeon isolated from the Dead Sea. Because of the importance of this work, Steitz shared the 2009 Nobel Prize in Chemistry with other scientists who made significant contributions to the understanding of ribosome structure.

Transfer RNA is the third main type of RNA and one of the smallest, usually only 70–90 nucleotides long. It carries the correct amino acid to the site of protein synthesis in the ribosome. It is the base pairing between the tRNA and mRNA that allows for the correct amino acid to be inserted in the polypeptide chain being synthesized. Any mutations in the tRNA or rRNA can result in global problems for the cell because both are necessary for proper protein synthesis

[A diagram of the 2-dimensional tRNA which is a single long strand of RNA folded into a plus shape with loops on the sides and bottom. The regions where the tRNA is folded so that there are 2 parts of the strand forming the linear portions of the plus are held together by hydrogen bonds labeled intramolecular pairing. The loop at the bottom has a set of 3 letters that are complimentary to 3 letters on the mRNA. The top part of the plus has a single stranded end at the 3-prime end; this is attached to an amino acid. B) The 3-dimensional structure looks like single strand folded into a double stranded structure with a bend in the middle.]

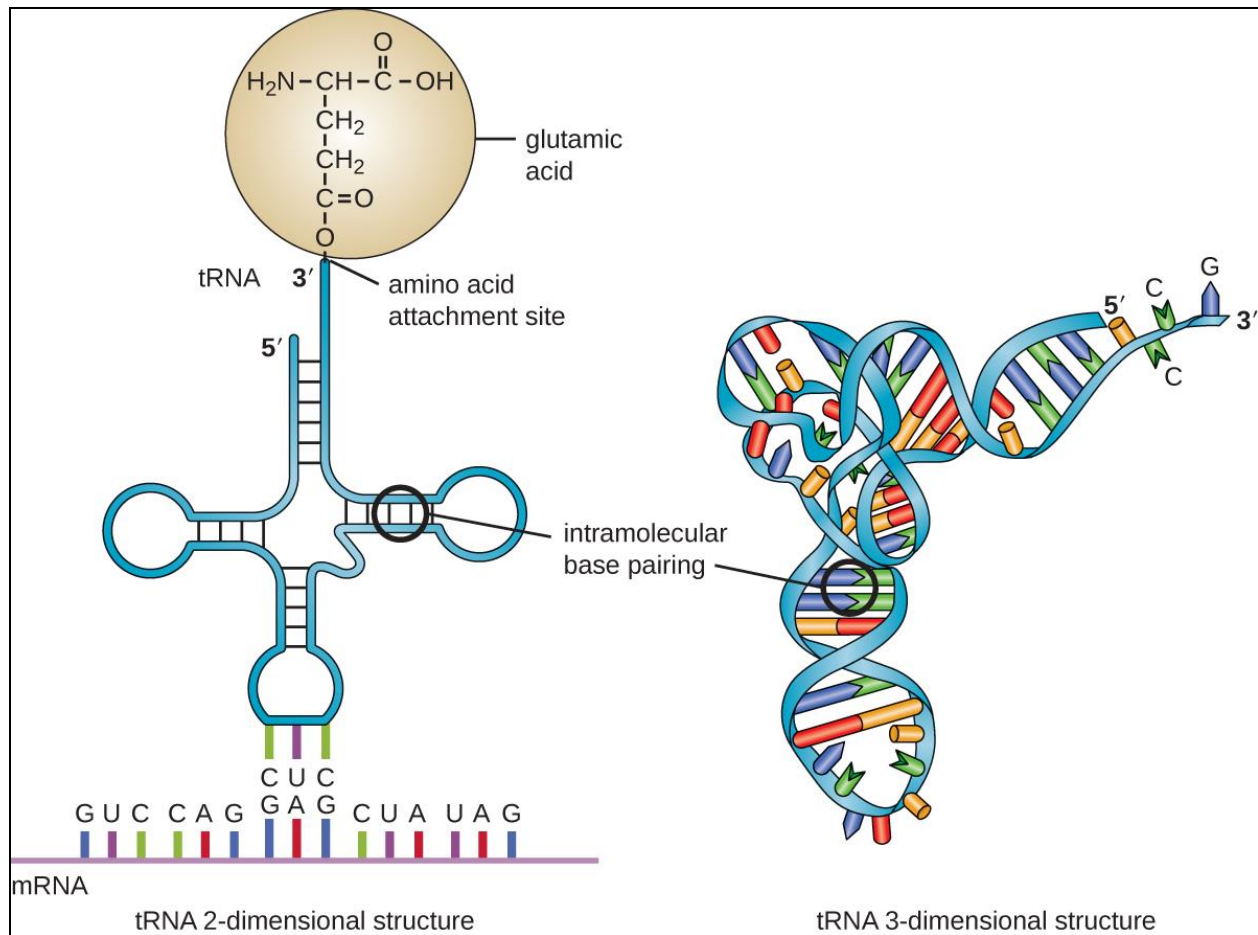
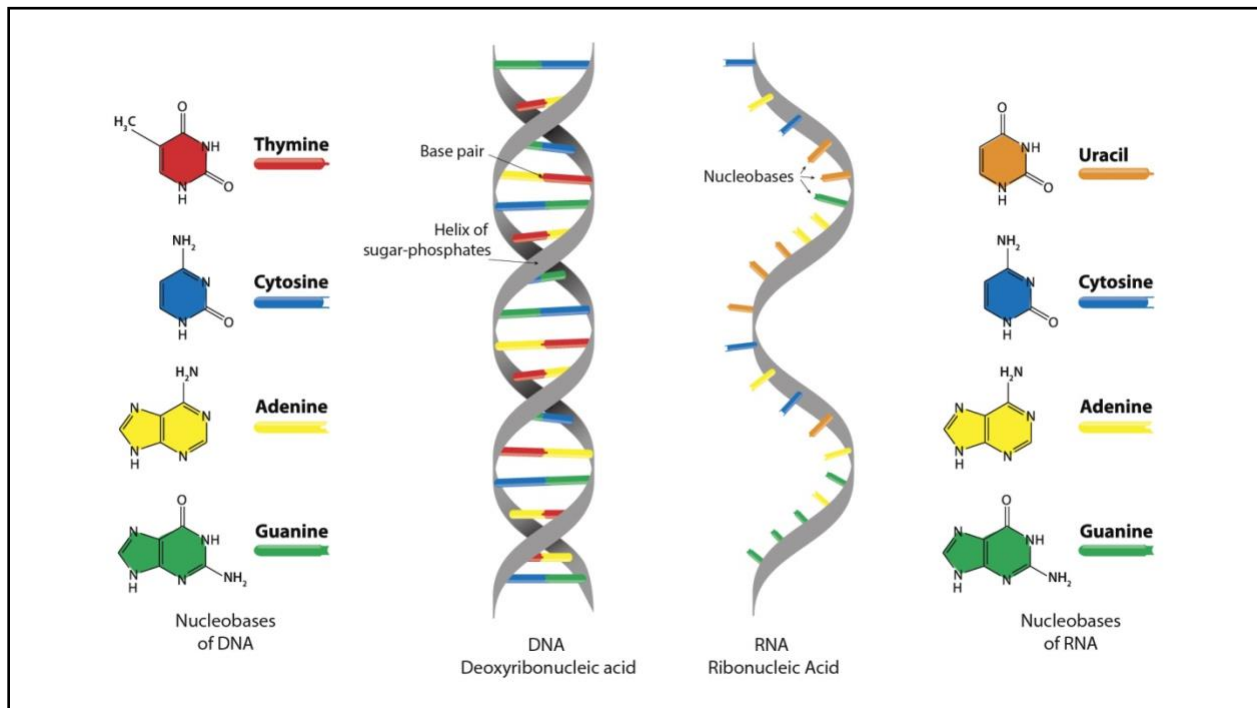
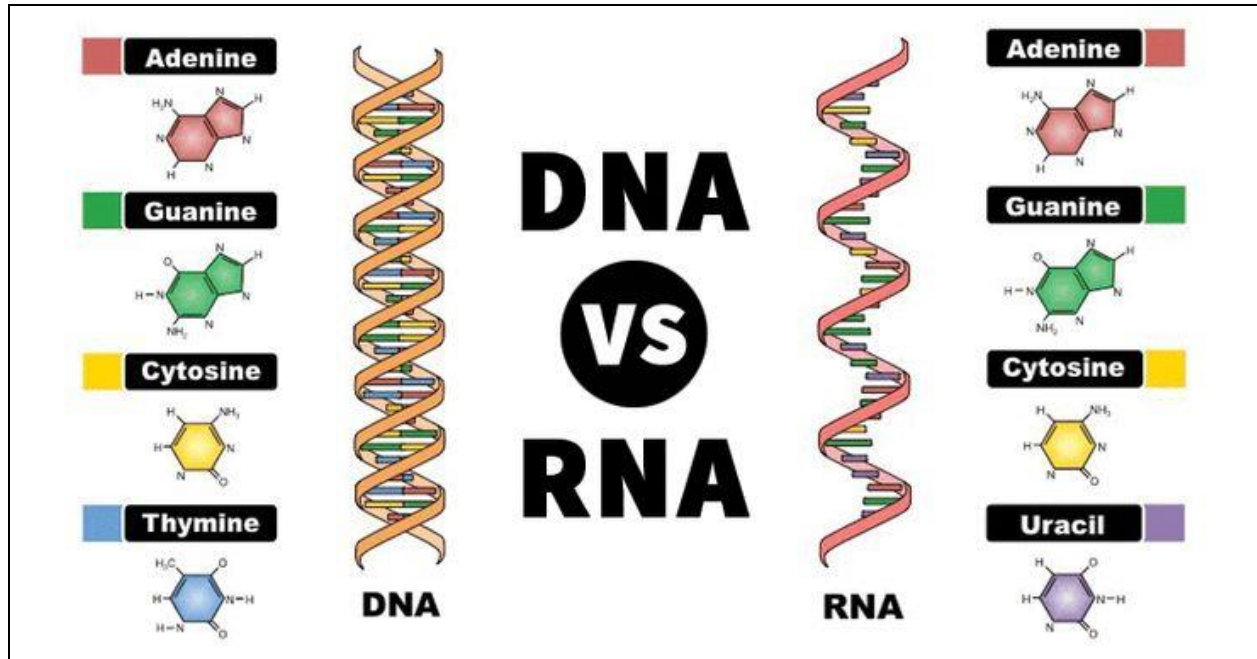


Fig: A tRNA molecule is a single-stranded molecule that exhibits significant intracellular base pairing, giving it its characteristic three-dimensional shape.

	mRNA	rRNA	tRNA
Structure	Short, unstable, single-stranded RNA corresponding to a gene encoded within DNA	Longer, stable RNA molecules composing 60% of ribosome's mass	Short (70-90 nucleotides), stable RNA with extensive intramolecular base pairing; contains an amino acid binding site and an mRNA binding site
Function	Serves as intermediary between DNA and protein; used by ribosome to direct synthesis of protein it encodes	Ensures the proper alignment of mRNA, tRNA, and ribosome during protein synthesis; catalyzes peptide bond formation between amino acids	Carries the correct amino acid to the site of protein synthesis in the ribosome

RNA as Hereditary Information

Although RNA does not serve as the hereditary information in most cells, RNA does hold this function for many viruses that do not contain DNA. Thus, RNA clearly does have the additional capacity to serve as genetic information. Although RNA is typically single stranded within cells, there is significant diversity in viruses. Rhinoviruses, which cause the common cold; influenza viruses; and the Ebola virus, are single-stranded RNA viruses. Rotaviruses, which cause severe gastroenteritis in children and other immune compromised individuals, are examples of double-stranded RNA viruses. Because double-stranded RNA is uncommon in eukaryotic cells, its presence serves as an indicator of viral infection. The implications for a virus having an RNA genome instead of a DNA genome are discussed in more detail in Viruses.



DNA vs. RNA – A comparison chart

Comparison	DNA	RNA
Full Name	Deoxyribonucleic Acid	Ribonucleic Acid
Function	DNA replicates and stores genetic information. It is a blueprint for all genetic information contained within an organism.	RNA converts the genetic information contained within DNA to a format used to build proteins, and then moves it to ribosomal protein factories.
Structure	DNA consists of two strands, arranged in a double helix. These strands are made up of subunits called nucleotides. Each nucleotide contains a phosphate, a 5-carbon sugar molecule and a nitrogenous base.	RNA only has one strand, but like DNA, is made up of nucleotides. RNA strands are shorter than DNA strands. RNA sometimes forms a secondary double helix structure, but only intermittently.
Length	DNA is a much longer polymer than RNA. A chromosome, for example, is a single, long DNA molecule, which would be several centimetres in length when unraveled.	RNA molecules are variable in length, but much shorter than long DNA polymers. A large RNA molecule might only be a few thousand base pairs long.
Sugar	The sugar in DNA is deoxyribose, which contains one less hydroxyl group than RNA's ribose.	RNA contains ribose sugar molecules, without the hydroxyl modifications of deoxyribose.
Bases	The bases in DNA are Adenine ('A'), Thymine ('T'), Guanine ('G') and Cytosine ('C').	RNA shares Adenine ('A'), Guanine ('G') and Cytosine ('C') with DNA, but contains Uracil ('U') rather than Thymine.

Base Pairs	Adenine and Thymine pair (A-T) Cytosine and Guanine pair (C-G)	Adenine and Uracil pair (A-U) Cytosine and Guanine pair (C-G)
Location	DNA is found in the nucleus, with a small amount of DNA also present in mitochondria.	RNA forms in the nucleolus, and then moves to specialized regions of the cytoplasm depending on the type of RNA formed.
Reactivity	Due to its deoxyribose sugar, which contains one less oxygen-containing hydroxyl group, DNA is a more stable molecule than RNA, which is useful for a molecule which has the task of keeping genetic information safe.	RNA, containing a ribose sugar, is more reactive than DNA and is not stable in alkaline conditions. RNA's larger helical grooves mean it is more easily subject to attack by enzymes.
Ultraviolet (UV) Sensitivity	DNA is vulnerable to damage by ultraviolet light.	RNA is more resistant to damage from UV light than DNA.

What are the key differences between DNA and RNA?

We can identify five key categories where DNA and RNA differ:

- Function
- Sugar

- Bases
- Structure
- Location

Function

DNA encodes all genetic information, and is the blueprint from which all biological life is created. And that's only in the short-term. In the long-term, [DNA is a storage device, a biological flash drive that allows the blueprint of life to be passed between generations.](#)

RNA functions as the reader that decodes this flash drive. This reading process is multi-step and there are specialized RNAs for each of these steps. Below, we look in more detail at the three most important types of RNA.

What are the three types of RNA?

Messenger RNA (mRNA) copies portions of genetic code; a process called transcription, and transports these copies to ribosomes, which are the cellular factories that facilitate the production of proteins from this code.

Transfer RNA (tRNA) is responsible for bringing amino acids, basic protein building blocks, to these protein factories, in response to the coded instructions introduced by the mRNA. This protein-building process is called translation.

Finally, **Ribosomal RNA (rRNA)** is a component of the ribosome factory itself, without which protein production would not occur.

Sugar

Both DNA and RNA are built with a sugar backbone, but whereas the sugar in DNA is called deoxyribose (left in image), the sugar in RNA is called simply ribose (right in image). The 'deoxy' prefix denotes that, whilst RNA has two hydroxyl (-OH) groups attached to its carbon backbone, DNA has only one, and has a lone hydrogen atom attached instead. RNA's extra hydroxyl group proves useful in the process of converting genetic code into mRNAs that can be made into proteins, whilst the deoxyribose sugar gives DNA more stability.

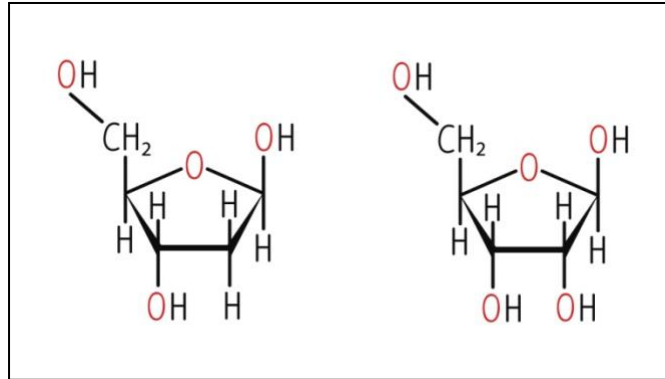


Figure: The chemical structures of deoxyribose and ribose sugars are compared. The chemical structures of deoxyribose (left) and ribose (right) sugars. **Bases**

The nitrogen bases in DNA are the basic units of genetic code, and their correct ordering and pairing is essential to biological function. The four bases that make up this code are adenine (A), thymine (T), guanine (G) and cytosine (C). Bases pair off together in a double helix structure, these pairs being A and T, and C and G. RNA doesn't contain thymine bases, replacing them with uracil bases (U), which pair to adenine.

Structure

While the ubiquity of Francis Crick and James Watson's (or should that be Rosalind Franklin's?) DNA double helix means that the two-stranded structure of DNA structure is common knowledge, RNA's single-stranded format is not as well known.

RNA can form into double-stranded structures, such as during translation, when mRNA and tRNA molecules pair. DNA polymers are also much longer than RNA polymers; the 2.3m long human genome consists of 46 chromosomes, each of which is a single, long DNA molecule. RNA molecules, by comparison, are much shorter.

Location

Eukaryotic cells, including all animal and plant cells, house the great majority of their DNA in the nucleus, where it exists in a tightly compressed form, called a chromosome⁴. This squeezed format means the DNA can be easily stored and transferred. In addition to

nuclear DNA, some DNA is present in energy-producing mitochondria, small organelles found free-floating in the cytoplasm, the area of the cell outside the nucleus.

The three types of RNA are found in different locations. mRNA is made in the nucleus, with each mRNA fragment copied from its relative piece of DNA, before leaving the nucleus and entering the cytoplasm. The fragments are then shuttled around the cell as needed, moved along by the cell's internal transport system, the cytoskeleton. tRNA, like mRNA, is a free-roaming molecule that moves around the cytoplasm. If it receives the correct signal from the ribosome, it will then hunt down amino acid subunits in the cytoplasm and bring them to the ribosome to be built into proteins⁵. rRNA, as previously mentioned, is found as part of ribosomes. Ribosomes are formed in an area of the nucleus called the nucleolus, before being exported to the cytoplasm, where some ribosomes float freely. Other cytoplasmic ribosomes are bound to the endoplasmic reticulum, a membranous structure that helps process proteins and export them from the cell.

Unusual types of DNA and RNA

The structure we have described in this article is certainly the most common form of DNA, but it isn't the whole story. Other forms of both DNA and RNA exist that subvert the classical structures of these nucleic acids.

Z-DNA

While the structure of DNA you will see above – and in any biology textbook you might care to open – has a right-handed helix, DNA molecules with left-handed helices also exist. These are known as Z-DNA. Canonical, “classic” DNA is called B-DNA.

Z-DNA molecules are:

Thinner (18 Å wide as opposed to 20 Å wide B-DNA)

Have a different repeating unit (two base pairs as opposed to one)

Have different twist angles between bases

Z-DNA is thought to play a role in regulating gene expression and may be produced in the wake of DNA processing enzymes, like DNA polymerase.

A-DNA

Identified at the same time as B-DNA by Rosalind Franklin, A-DNA is an alternative DNA structure that often appears when the molecule is dehydrated. Many crystal structures of DNA are in an A-DNA form. It has a shorter structure, with different numbers of base pairs per turn and tilt than B-DNA. A-DNA's biological relevance has been greatly expanded on in recent years, and it is now recognized that A-DNA is involved in many roles, such as:

Binding to DNA enzymes, such as polymerases – this transition may enable specific atoms to be exposed for enzymatic action.

Protection from damage – A-DNA is far less susceptible to ultraviolet ray damage, and spore-forming bacteria have been shown to adopt an A-DNA conformation, which may be a protective change.

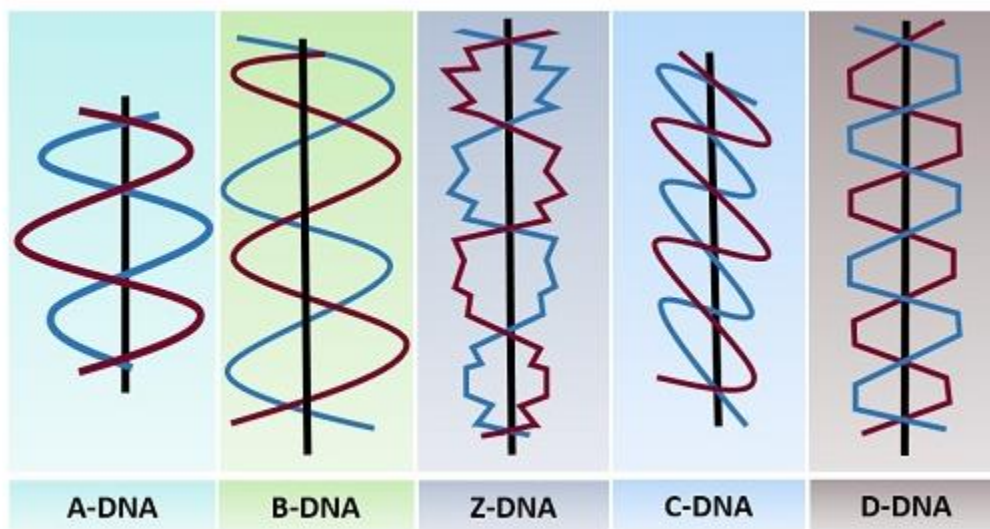
Triplex DNA

A triple-helix DNA structure can form when certain nucleobases – pyrimidine or purine – occupy the major grooves in conventional B-DNA. This can happen naturally or as part of intentional DNA-modifying strategies for research purposes.

Triplex-forming oligonucleotides (TFOs) can bind conventional two-stranded DNA, which can help guide agents that are used to modify DNA to specific genomic locations. H-DNA is an endogenous, triple-stranded DNA molecule that encourages mutation of the genome.

Table 7.2 Properties of major forms of DNA

Particulars	A DNA	B DNA	Z DNA
Helix	Right handed	Right handed	Left handed
Base pairs per turn	~11	~10.5	~12
Helical Diameter (nm)	2.6	2.0	1.8
Helical length (nm)	2.6	3.4	3.7
Shape	Broadest	Intermediate	Narrowest
Major Groove	Wide, deep	Narrow, deep	Flat
Minor Groove	Narrow, shallow	Broad, shallow	Narrow, deep

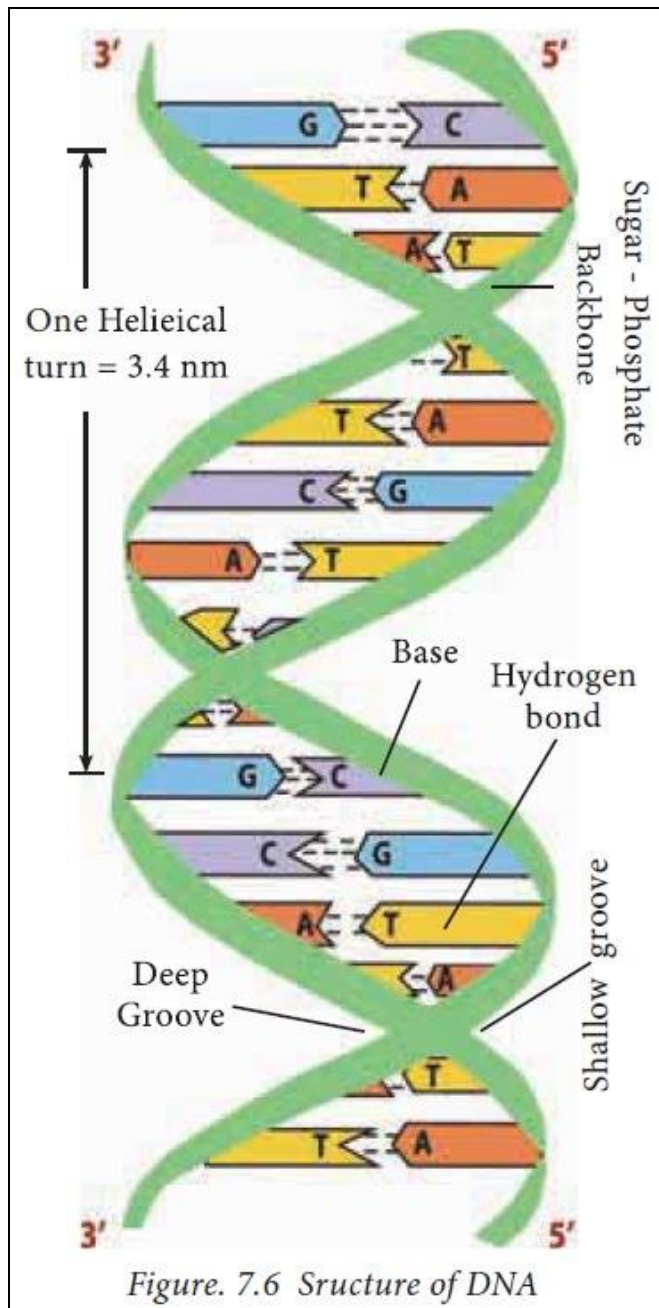


2. Salient features of the structure of DNA

The B form DNA, also known as the Watson- Crick DNA is the most stable and prevalent form of DNA. The important structural features of B- DNA are:

1. There are two polynucleotide chains in the DNA spirally twisted around each other to form a right handed double helix.

2. The sugar-phosphate backbones remain on the outside, while the core of the helix contains the purine and pyrimidine bases.
3. The diameter of DNA is 2 nm or 20 Å. The length of a complete turn of helix is 3.4 nm or 34 Å i.e. there are ~10.5bp per turn.
4. The DNA helix has a shallow groove called minor groove (-1.2nm) and a deep groove called major groove (-2.2nm). Proteins interact with DNA through the minor and major grooves without disrupting the DNA strands.
5. Each polynucleotide chain is made up of four different bases. The purine bases present in DNA are adenine and guanine and the pyrimidine bases present are thymine and cytosine. The sequence of purine and pyrimidine carry the genetic information whereas the sugar and phosphate groups perform the structural role.
6. Each polynucleotide chain has direction or polarity. Further, each polynucleotide chain has 5' phosphorylated and 3' hydroxyl ends.
7. The two strands run in opposite direction (i.e.) they are antiparallel.



dsRNA

Double-stranded RNA (dsRNA) is most commonly found as the genomic basis of many plant, animal and human viruses. These include Reoviridae and the rotaviruses, which are responsible for diseases like gastroenteritis. dsRNA molecules are potent immunogens – they activate the immune system, which then cuts the dsDNA as a

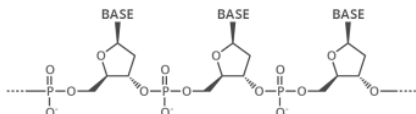
protective mechanism. The discovery of the protein machinery that permits this reaction led to the development of gene-silencing RNAi technology, which won the 2006 Nobel Prize for Physiology or Medicine.

		Second letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

What makes up the chemical structure of DNA?

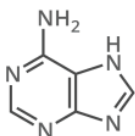


The sugar-phosphate backbone

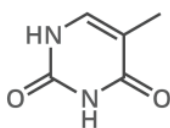


DNA is a polymer made up of units called nucleotides. The nucleotides are made of three different components: a sugar group, a phosphate group, and a base. There are four different bases: adenine, thymine, guanine and cytosine.

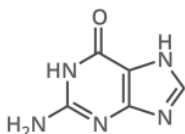
A Adenine



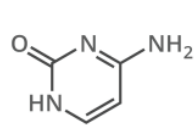
T Thymine



G Guanine

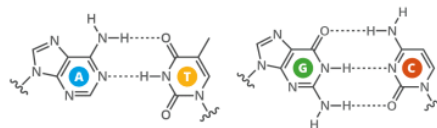


C Cytosine



What holds DNA strands together?

DNA strands are held together by hydrogen bonds between bases on adjacent strands. Adenine (A) always pairs with thymine (T), while guanine (G) always pairs with cytosine (C). Adenine pairs with uracil (U) in RNA.



From DNA to proteins

The bases on a single strand of DNA act as a code. The letters form three letter codons, which code for amino acids - the building blocks of proteins.



An enzyme, RNA polymerase, transcribes DNA into mRNA (messenger ribonucleic acid). It splits apart the two strands that form the double helix, then reads a strand and copies the sequence of nucleotides. The only difference between the RNA and the original DNA is that in the place of thymine (T), another base with a similar structure is used: uracil (U).



In multicellular organisms, the mRNA carries genetic code out of the cell nucleus, to the cytoplasm. Here, protein synthesis takes place. 'Translation' is the process of turning the mRNA's 'code' into proteins. Molecules called ribosomes carry out this process, building up proteins from the amino acids coded for.

